

Mobile Robot Primitive Mapping Using Machine Learning - Project Proposal

Group Elements

- Ana Couto, nº 200802466
- I. Antónia Brito, nº 202107271
- Pedro Sousa, nº 202108383

Goals

The objective of this project is to develop a machine learning (ML) model capable of recognizing primitive shapes within the surroundings of a mobile robot. To accomplish this, an E-puck robot equipped with LiDAR sensors will be utilized. The environment in which the robot operates will contain various obstacles, with walls structured as 2D primitives such as circles, rectangles, triangles, ellipses, and rhombuses. It is assumed that the maps do not include randomly shaped objects.

Upon gathering and processing data from the LiDAR sensor, a ML model, potentially a Deep Neural Network (DNN), will be trained. Once fully trained, this model should accurately output the pose (position and orientation) and parameters of the identified primitives. Each primitive's position will be defined by its centroid and additional parameters specific to the primitive type:

- Sphere: Radius.
- Rectangle: Width, height.
- Triangle: Relative position of the three vertices in relation to the centroid.
- Ellipse: Radii along both axes.

Two experiments will be conducted, one in which multiple LiDAR readings will be combined to theoretically achieve more accurate predictions. And a second experiment, involving stress testing the model against noisy input from the LiDAR to ensure robustness and reliability.

Proposed Approach

Robot: 'e-puck'

Sensors: GPS, Compass, LiDAR

Dataset construction: As first step, multiple maps will be created, varying in number and type of primitives included. After storing the ground truth values of the parameters of each primitive, we shall then move on into the LiDAR mapping, using the 'e-puck' robot.

The initial raw data that will be used to construct the final dataset, will consist of a series of LiDAR scans from multiple perspectives of different primitive-shaped objects. The scans will be one layered with an horizontal resolution of 512 rays and a field of view of 1.67 rad. This time, values like robot position and orientation should be stored as well as the point clouds formed from the readings.

As an alternative approach, we consider using a mapping algorithm derived from the popular SLAM (Simultaneous Localization and Mapping) method in order to create complete point cloud representations of the environment.

Model: As for the ML model, we will take inspiration from the PointNet Deep Neural Network [5], which was originally created to solve complex classification and segmentation problems of objects using 3D point clouds. Since this study will consist of a 2D primitive identification, we can adapt this method to best fit our goals.

Proposed Experiments

Experiment 1:

This experiment consists of combining multiple LiDAR scans into a singular point cloud. By doing this we aim to achieve better performance from our model, as well as being able to interpret more complex maps (with a variety of different primitives).

Experiment 2:

The second experiment consists of the gradual introduction of noise, either Gaussian noise into the dataset, or by adding noise directly to the LiDAR scans with a lower resolution sensor. We plan on experimenting with three different levels: low, medium and high. With this, we aim to understand the robustness of the model regarding the fluctuations of the LiDAR readings.

Evaluation Metrics

Precision: proportion of correctly identified primitives;

IoU (Intersection over Union): overlap between the ground truth and predicted bounding boxes of the primitives. It is particularly relevant when evaluating the accuracy of pose estimation. $\text{IoU} = \text{Area of Intersection} / \text{Area of Union}$ [1];

Mean Average Precision (mAP): average precision across all classes, evaluation of the model's performance on different primitive shapes [2];

Average root-mean-squared-error (RMSE): average root mean squared difference between the estimated parameters and ground-truth parameters [4];

Execution Time: time (seconds) taken by the model to process LiDAR data and make predictions.

Empirical Evaluation

To evaluate the best models, each experiment will be run 10 times and their evaluation metrics will be averaged. The following null hypothesis will be tested:

- Experiment 1:
H0: Single LiDAR readings trained model = Multiple LiDAR readings trained model
- Experiment 2:
H0: Noisy LiDAR readings = Clean LiDAR readings

Objectives (weekly)

Week0: April 1st - April 7th

- Set robot and virtual environment
- Build maps

Week1: April 8th - April 14th

- Run robot to collect LiDAR data scans (run scans with multiple values of noise)
- Create and store point clouds

Week2: April 15th- April 21st

- Clean and annotate datasets

Week3: April 22nd - April 28th

- Finish final dataset
- Start modeling phase

Week4: April 29th - May 5th

- Modeling
- Training

Week5: May 6th - May 12th

- Empirical evaluation

May 14th - Intermediate checkpoint

Objectives (intermediate delivery)

With the upcoming delivery date June 3rd, only 2 weeks from the end of week 5, we should have gathered all necessary data and therefore should shift our focus to the results interpretations and writing the scientific paper.

References

[1] Rezatofighi, H., Tsoi, N., Gwak, J., Sadeghian, A., Reid, I., & Savarese, S. (2019). Generalized intersection over union: A metric and a loss for bounding box regression. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (658-666). <https://doi.org/10.1109/CVPR.2019.00075>

[2] [Breaking Down Mean Average Precision \(mAP\) | by Ren Jie Tan | Towards Data Science](#)
(29-3-2024)

[3] Xia, S., Chen, D., Wang, R., Li, J., & Zhang, X. (2020). Geometric Primitives in LiDAR Point Clouds: A Review, in IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 13, 685-707. <https://doi.org/10.1109/JSTARS.2020.2969119>

[4] Xu, N., Qin, R. & Song, S. (2023) Point cloud registration for LiDAR and photogrammetric data: A critical synthesis and performance analysis on classic and deep learning algorithms, ISPRS Open Journal of Photogrammetry and Remote Sensing, 8, 100032. <https://doi.org/10.1016/j.ophoto.2023.100032>

[5] Zheng, S. & Castellani, M.(2023). Primitive shape recognition from real-life scenes using the PointNet deep neural network. Int J Adv Manuf Technol 124, 3067–3082. <https://doi.org/10.1007/s00170-022-09791-z>